The University of Georgia

ELEE 4230/6230: Sensors and Transducers
Course Project – Fall 2025

Reaction-Time Training Button

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Abstract

Reaction time plays an important role in activities that require rapid decision making, including racing, gaming, driving, and general safety in everyday life. Because of this, a device that measures and helps improve reaction speed can be both useful and engaging. This project aimed to design a compact, self-contained reaction time training system that allows users to practice responding to a visual cue while integrating multiple raw sensors, embedded signal processing, and an intuitive user interface. The system used a force sensitive resistor and an infrared reflective sensor as dual input methods to detect a user's response, while a thermistor circuit monitored temperature to ensure safe operation. The Raspberry Pi Pico W handled all signal conditioning and processing, including a Kalman filter for smoothing the FSR readings. An OLED display provided real-time instructions and feedback, and the system also included a feature that uploaded reaction times to ThingSpeak and automatically retrieved the top three scores to display to a live leaderboard.

Testing showed that the system operated reliably, measured reaction times within expected human ranges, and performed well during the in-class demonstration. The OLED interface clearly presented game states and results, and IoT integration successfully posted and retrieved leaderboard data when WiFi was available. Overall, the project met its goals by producing a functional and engaging reaction time trainer that demonstrates effective use of analog sensing, embedded programming, and wireless data integration. Future improvements could target PCB fabrication, single battery operation, and speeding up leaderboard refresh, but the current design offers a solid and fully working version one.

Problem Introduction

Reaction time affects almost every part of daily life, even if people rarely think about it directly. Whether someone is braking during driving, reacting to the start of a race, avoiding a moving object, or responding to visual cues in competitive gaming, the speed at which the brain perceives a stimulus and the muscles initiate a response can make a meaningful difference in safety and performance. I have always been interested in how people train this kind of rapid decision making, so I wanted to explore the underlying sensing and measurement challenges involved in capturing human reaction time accurately.

The central problem of this project is how to quantify a user's reaction speed to a controlled stimulus in a reliable, repeatable, and measurable way. To do this, several physical quantities must be sensed with good fidelity. First, the moment a user initiates a physical response must be detected. This involves measuring force or pressure changes applied by the user. Second, a secondary sensing method must confirm that the user's hand or object truly moves when responding to the stimulus. Third, the device must ensure safe operation by monitoring environmental or component temperatures to avoid overheating. Each of these quantities requires a different type of transduction: resistive sensing for force, optical sensing for presence/motion, and temperature-dependent resistance for heat measurement.

These measurement tasks introduce several classical sensing challenges. For example, force sensors like force-sensitive resistors (FSRs) experience loading effects, meaning that any interface circuit adds its own impedance and may shift the measured voltage if not designed with sufficiently high input impedance. Similarly, optical reflective sensors depend on surface reflectivity and object distance, creating non-idealities such as threshold uncertainty and false detections. Even temperature monitoring with a thermistor involves non-linear resistance-to-temperature relationships, requiring proper modeling to avoid large conversion errors. All of these sensors must be conditioned, sampled, and interpreted in a way that preserves timing accuracy, since even small timing delays or noise could corrupt the measurement of reaction time.

At its core, the problem is not simply to create a "game," but to establish a measurement platform that captures the user's response with accuracy and repeatability. Reaction time is fundamentally a latency measurement between a visual cue and a measurable human-generated signal. The challenge lies in ensuring that the sensors detect the user's response at the precise moment it occurs, while minimizing delays introduced by sensor physics, circuit loading, signal conditioning, or environmental influences. By clearly defining what needs to be measured and by understanding how different sensors behave in real-world conditions, this project frames reaction-time training as a sensing and measurement problem rather than just a user-interface task.

Figure: AI Generated Representation of Reaction Time Applications.

Physical Environment and Project Constraints

Course-Mandated Constraints:

- Raw Sensor Requirement
  - The force-sensitive resistor (FSR) along with the thermistor were purely analog sensors. Both sensors required appropriate analog filtering and explicit calibration curves to translate voltage into force and temperature.
- Active Signal Processing Requirement:
  - The thermistor signal was actively amplified by the INA121, which referenced a 1.65 volt offset and used a calibrated gain of approximately six.
  - The FSR signal used active smoothing via a Kalman filter and an analog low-pass filter whose output connected to a voltage follower to stabilize the ADC readings.
- Analog Filtering Requirement:
  - The FSR sensing circuit included a low-pass RC filter to remove rapid noise spikes inherent in resistive sensors. The thermistor's INA121 circuit has bandwidth limitations and internal filtering that reduce high-frequency noise, functioning as part of the overall analog filtering strategy.
- Battery Powered Requirement:
  - In this case, a single battery was not feasible because WIFI transmission on the Pico 2W caused voltage drops that interfered with sensor readings. Trudgen approved the use of two batteries, one dedicated to the 5 volt regulator feeding VSYS and one dedicated to powering the 3.3 volt analog circuitry.
- Algorithm Requirements:
  - The firmware includes all required algorithms:
    - Each sensor is encapsulated in its own dedicated subfunction that returns its measurement.
    - A Kalman filter is applied to the FSR's voltage signal.
    - The "go beyond" requirement is met through full IoT integration with ThingSpeak, including data posting, retrieval, and real-time leaderboard displayed on the OLED screen.
- Physics Requirements:
  - The system includes redundancy from two different branches of physics for the primary measurement (reaction time detection):
    - Mechanical force detection: FSR (mechanical resistance change).
    - Optical detection: infrared reflective sensor (photon reflection).
  - This dual-sensor redundancy ensures robust event detection even if a single sensor has loading or environmental error.
  - Temperature measurement adds a third environmental parameter, meeting the requirement for at least three total physical measurements

Practical Project Constraints

- Time Constraints:
  - The system had to be fully designed, built, debugged, and demonstrated before the end of the semester. This required fast prototyping and early sensor testing so that any problematic components could be replaced without delaying integration. Firmware development also had to be staged carefully, beginning with sensor validation and ending with WiFi, ThingSpeak IoT functionality, and the final reaction-time logic.
- Budget and Component Availability:
  - Only essential components were purchased, and the bill of materials (BOM) was kept as compact as possible **(A complete bill of materials is included in Appendix A).**
  - Availability also played a role. Since some sensors can have long shipping times or unpredictable stock, parts were selected that had reliable, widely available datasheets and fast shipping.
- Breadboard Space Constraints
  - The physical prototyping area was small. This placed constraints on: The number of support components, the choice of switching topology for powering regulators, the layout of the analog amplifier and sensor networks, and the selection of discrete vs. integrated hardware. This is why certain simplifications, such as using a single P-MOSFET for battery switching, were important.

Physical Environment

The environment in which the device is intended to operate is a typical indoor academic environment, such as a classroom or lab in Driftmier Engineering Center. Because this is a human-interaction device, its environment is characterized by moderate temperatures, stable humidity, and predictable lighting. However, several environmental conditions must still be considered when evaluating sensor performance

- Temperature: Indoor, approximately 65-75 F (although the device does work within a wider non-harsh temperature range). Temperature measurement circuitry must operate correctly within this range and detect an over-temperature condition if the electronics themselves heat up.
- Ambient Lighting: The IR Sensor must tolerate variations caused by overhead lighting, reflections, and shadows. Indoor lighting is not intense enough to saturate the IR receiver, but ambient IR noise still exists.
- Mechanical Loading: The FSR must operate under the force ranges applied by human fingertips or hands. Loading errors are possible because force-sensitive resistors inherently add series resistance and exhibit nonlinearities.

Loading Errors and Measurement Considerations:

- The FSR slightly loads the measurement node because its resistance is part of a voltage divider. This creates sensitivity to battery voltage fluctuations and component tolerance.
- The thermistor is part of a resistive network and therefore loads the excitation bridge. Temperature readings depend on resistor tolerance and amplifier gain accuracy.
- The IR sensor is affected by object distance, reflectivity, and alignment. Although this is not destructive, it introduces variability in detection time.

What Was Modeled: Expected force ranges, expected temperature range, indoor lighting level, sensor dynamic ranges, and battery voltage sag under digital load.

What Was Not Fully Modeled: Humidity effects on FSR compressibility, Long-term thermal drift of the INA121, EMI effects on the analog lines, and rare false triggers from IR reflections in bright environments.

These unmodeled effects are not expected to impact the project significantly but are acknowledged as possible sources of error. **A filled out "General Safety Checklist For Sensors" can be viewed in** **Appendix B**.

Sensor Selection

A reaction-time trainer needs to reliably detect two things:

1. The exact moment the user intends to respond, and
2. The environmental conditions that could affect sensor performance or user safety.

Because of these constraints, the sensor selection process began broadly. For each type of measurement the trainer needed mechanical force/contact, optical confirmation, and temperature. Multiple candidate sensors were considered and after evaluating pros/cons in the context of this project's constraints and physical environment, a set of three sensors emerged as the best fit.

Detecting User Press / Reaction Event

Sensor Type	**Pros**	**Cons**
Mechanical pushbutton	Simple, cheap, digital output	Poor repeatability, bounce, no force measurement, no raw analog signal (violates raw-sensor requirement)
Force Sensitive Resistor (FSR)	Analog raw signal, easy to integrate with RC + active filtering, confirms to finger press.	Nonlinear, temperature-dependent, required calibration

Strain gauge + instrumentation amp	Very accurate, highly linear	Expensive, requires precision mounting, overkill for reaction-time taps
Piezo disk	Very fast response	Highly sensitive to vibration, requires specialized conditioning

Final Choice: Force Sensitive Resistor (FSR)
The FSR was selected because it satisfies the raw-sensor requirement, provides a real analog signal that requires active processing and filtering, and naturally matches the physical action of the user's finger pressing a button. Its compact size is also a plus, along with its speed being far beyond what is necessary for human reaction timing. The nonlinearity and drift are acceptable because absolute force accuracy is not central to the problem; only the moment of actuation and peak-force estimation matter.

Redundant Reaction Detection (Second Branch of Physics)

Sensor Type	**Pros**	**Cons**
Capacitive touch sensor	Easy to use; touch-based	Not a raw sensor; strongly susceptible to environment noise.
IR reflective sensor	Uses light instead of pressure, fast; independent physics	Requires tuning; sensitive to alignment
Ultrasonic sensor	Robust detection	Slow; excessive for short-range detection
Hall-effect switch	Clean digital output	Requires magnets attached to button assembly

Final Choice: IR Reflective Sensor
This was chosen because it satisfies the redundancy requirement using a completely different branch of physics, it is extremely fast, and provides a digital event signal that pairs well with the FSR. It is unaffected by small vibrations, safe for the user, and operates completely within indoor conditions without RF or lighting concerns. Its onboard comparator makes detection robust to noise while still letting the project implement custom filtering logic in software.

Temperature Monitoring (Safety Requirement)

Sensor Type	Pros	Cons
Digital temperature IC	Accurate; easy to read	Not a raw sensor; does not satisfy analog processing requirement
Thermistor + resistor divider	Cheap and raw; simple	Requires calibration; nonlinear
Thermocouple	Very large temperature range	Requires specialized amplifiers; overkill
RTD	High accuracy	Requires precise current source and instrumentation

Final Choice: NTC Thermistor
A NTC thermistor in a bridge configuration with a INA121 in-amp satisfies the raw-sensor requirement and provides an ideal analog signal for temperature cutoff logic. The thermistor is inexpensive, easy to mount, and operates well in the expected environment. The amplifier provides the gain, level shifting, and noise rejection needed for the Pico's ADC, fulfilling the constraint of using meaningful active analog processing.

Final Sensor Choices

Measurement	Sensor	Physics Domain	Why this was the best choice
Primary Reaction Event	FSR (raw analog)	Mechanical resistance change	Raw sensor requirement; compact; safe to touch; fast; works with analog filtering + Kalman algorithm
Redundant reaction event	IR reflective module	Infrared optical reflection	Second branch of physics; fast; robust; provides clean digital edge timing
Temperature	NTC Thermistor	Thermistor resistance vs temperature	Raw; non-linear; allows active filtering and modeling; helps prevent over-temperature

			behavior

Raw Sensor Dynamics and Signal Processing

Force-Sensitive Resistor (FSR)

Note: The steps for characterizing the FSR can be found in Appendix C

- Raw Physics
    - The force-sensitive resistor (FSR) is a polymer-based resistive sensor whose sensitive element is a pressure-dependent conductive ink. When force is applied, conductive particles inside the polymer are pressed closer together, decreasing resistance. The response is nonlinear, with a steep resistance drop at low forces and a gradual slope toward saturation.
    - The key characteristics of the raw physics are:
        - Monotonic relationship: more force -> lower resistance
        - Nonlinear; strongest sensitivity below ~1.5 N
        - Exhibits hysteresis and creep due to polymer behavior
        - Saturates at higher forces
        - Sensitive to contact area and finger placement
    - A simple Force vs Voltage plot (from characterization) shows this shape clearly: fast rise at low loads, mid-range quasi-linear region, and a high-force plateau:

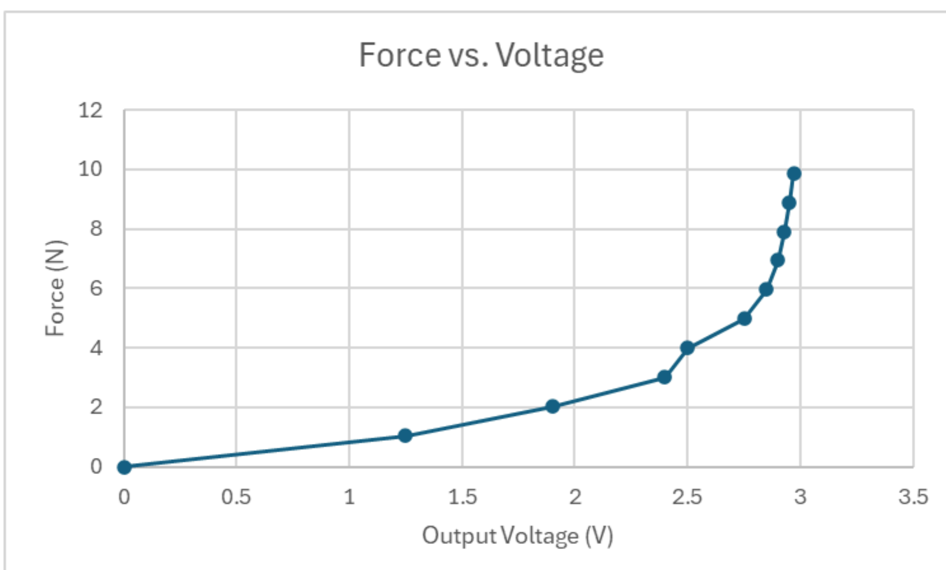


Figure: Plot showing Force vs. Voltage.

- Mathematic Model Chosen (pros/cons)

  - Because the raw FSR response is nonlinear and inconsistent across individual units, the project uses a piecewise-linear interpolation model built directly from calibration data. Each voltage breakpoint is matched to a measured force value, and the force for intermediate voltages is computed by linear interpolation.
  - Why this model?
    - Computes extremely fast on a microcontroller
    - Reflects the actual measured response rather than an assumed theoretical one
    - Accurate within the operating range
    - Easy to expand by adding more calibration points
  - What the model does NOT include:
    - Hysteresis between loading and unloading
    - Temperature variation
    - Time-based creep
    - Multi-dimensional effects like finger area or angle
    - Behavior above the saturation voltage (clamped at 2.9 V)
  - Pros:
    - Simple, robust, and inexpensive computationally
    - Matches real data well
    - Works directly with the filtered ADC voltage
  - Cons:
    - Limited accuracy outside the measured range
    - Ignores dynamic material behavior
    - Requires calibration to be meaningful
  - Model Range:
    - Valid from 0 to 2.9 V (~0-6.93 N). Above that, output is clamped at the maximum calibrated force.
- Transducers
  - The FSR itself is the transducer, It directly converts force into resistance. There is no intermediate transducer stage. The sensor is placed in a voltage divider with a fixed resistor, converting resistance into a measurable voltage. This voltage is then conditioned and sent to the ADC of the Pico.
  - Because the FSR is an inherently passive device, all dynamic behavior (filtering, amplification, smoothing) must be handled in the signal-conditioning and digital-processing chain.
- Signal Processing Options
  - Option 1: Simple Voltage Divider
    - Lowest cost and simplest wiring
    - Directly compatible with the Pico 2W ADC
    - Susceptible to noise and momentary spikes during fast presses

        - Requires digital smoothing for stable readings
    - Option 2: Voltage Divider + Passive RC Low-Pass Filter
        - Removes high-frequency noise and jitter
        - Minimal delay
        - Helps stabilize readings during fast finger taps
        - Good tradeoff between responsiveness and smoothness
    - Option 3: Active Signal Conditioning (Op-Amp Buffer or Amplifier)
        - Would offer more stable impedance to the ADC
        - Increases PCB space and power consumption
        - Not necessary because the divider already provides sufficient voltage range
    - Option 4: Post-ADC Digital Filtering Only
        - No added components
        - Does not prevent noise entering the ADC stage
        - Works best when combined with Kalman filtering but still vulnerable to fast spikes
- Power Options
    - The FSR requires only milliwatts, but the overall system imposed power-related constraints.
    - Option 1: Single 9V battery through 3.3V regulator
        - Simple
        - Insufficient when WiFi is active on the Pico
        - Voltage sag affected ADC stability
    - Option 2: Two Battery System through 3.3 V and 5 V regulators. (Final Choice)
        - One 9V for the 3.3V rail
        - One 9V for the 5V rail
        - Approved by the instructor due to WiFi current spikes
        - Provided stable voltage readings for the FSR ADC
    - Option 3: USB power
        - Stable but not allowed for the final project because of required battery operation
- Final Signal Processing Chosen
    - The chosen signal-conditioning circuit consisted of:
    - 1. FSR in a voltage divider
        - FSR as the top resistor
        - Fixed resistor chosen to maximize sensitivity in the typical press range
    - 2. Op-amp voltage follower (rail-to-rail buffer)
        - Prevents divider impedance from loading the ADC input
        - Provides a stable, low-impedance drive for the Pico ADC
    - 3. Passive RC Low-Pass Filter

        - Removes hand tremor noise and electrical interference
        - Helps prevent aliasing before sampling
        - Cutoff chosen around tens of Hz to match human reaction-time dynamics
    - 4. Pico ADC Sampling + Oversampling
        - Averaging multiple ADC reads improves resolution.
    - 5. Kalman Filtering and Calibration
        - Performed in code after digitization.
        - Converts raw voltage into force using the piecewise-linear model.

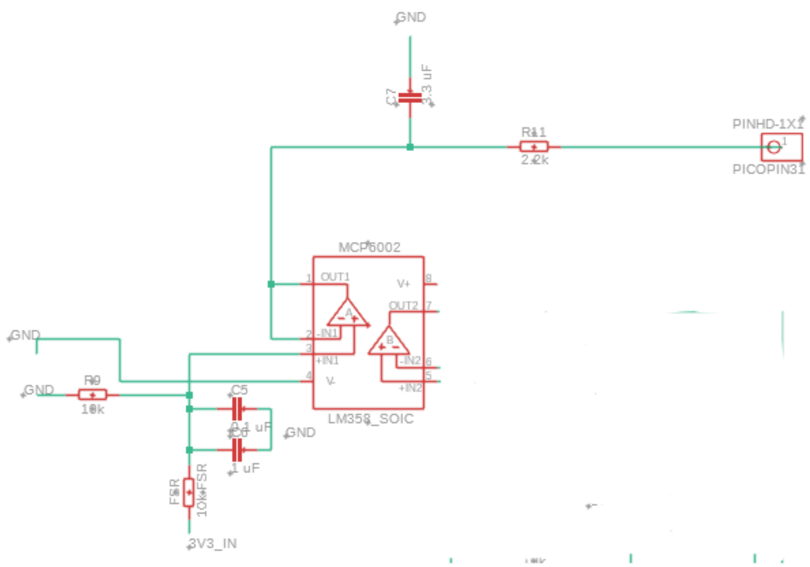


Figure: Circuit Schematic for FSR.

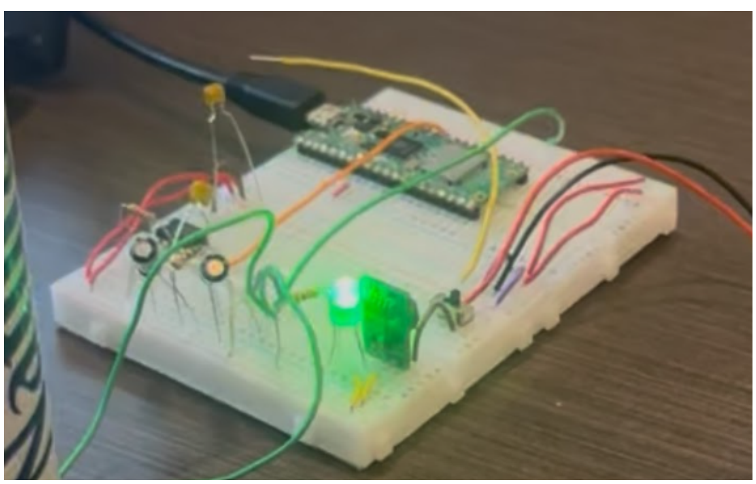


Figure: Physical Circuit For FSR.

- Filtering Chosen

- Analog Filtering: A first-order RC low-pass filter with a cutoff between 10 and 20 Hz.
    - R = 2.2k ohms; absorbs high-frequency jitter.
    - C = 3.3 uF; stores short bursts and smooths them
- Digital Filtering:
    - Oversampling (8 readings)
    - A 1D Kalman filter with Q = 5e-5 and R = 2e-4
- Why this filter was chosen:
    - Human reaction events occur at low frequencies
    - Filtering should not add measurable latency
    - The Kalman filter suppresses noise while preserving fast rising edges

- Microcontroller Code
    - The FSR is implemented as a subfunction that performs the entire signal-processing pipeline:
    - 1. *read_voltage()*
        - Performs oversampled ADC readings (8 samples)
        - Converts raw counts to voltage using Vref = 3.3 V
    - 2. Kalman Filter
        - Smooths the voltage while preserving rising-edge timing
        - State update:
            - Predict: x = x
            - Update: x = x + K(z-x)
        - Tuned parameters Q and R determine responsiveness
    - 3. Calibration
        - The filtered voltage is passed into *force_from_voltage_linear()*
        - This function uses piecewise-linear interpolation based on measured calibration data.
        - Values above the saturation threshold are clamped at 6.93 N
    - 4. Resolution
        - Effective ADC resolution after oversampling is approximately 11.5 to 12 bits.
        - Force resolution in the most sensitive region is roughly 0.05 to 0.1 N
    - A script containing code for just the FSR can be viewed through the link below. *(Note: This was incorporated in the final system through its own function call)* https://drive.google.com/file/d/1U8uFpUjVGs97kpLm627riwAQdn-xRBHW/view?usp=sharing
- Kalman Filter Used
    - The Kalman filter used for the FSR is a one-dimensional state estimator designed to smooth the voltage measurement while preserving timing accuracy. The state variable is the true sensor voltage. The measurement is the raw ADC voltage.

- Algorithm Steps:

1. Predict
    a. x_pred = x
    b. P_pred = P + Q
2. Update
    a. K = P_pred / (P_pred + R)
    b. x = x_pred + K (z - x_pred)
    c. P = (1 - K) P_pred

- Where:
    - x is the voltage estimate
    - P is the estimate variance
    - Q is the process noise (responsiveness)
    - R is the measurement noise (smoothness)
    - z is the new ADC measurement
- This filter improves stability substantially without introducing noticeable delay, making it ideal for capturing the moment the FSR is first pressed in a reaction-time trial.

- Video Results
    - A quick demo showing the FSR working in isolation can be viewed through the link below: https://drive.google.com/file/d/1MrxEfP5ZJYySobI3LeGNIgJOROD_R5Yg/view?usp=sharing

Thermistor

- Raw Physics
    - The thermistor used in the project is a negative temperature coefficient (NTC) device. The sensitive element is a semiconductor bead whose resistance decreases as temperature increases. The underlying physics is based on charge carrier activation: as temperature rises, additional carriers become available, decreasing resistivity.
    - The response is strongly nonlinear and follows the typical exponential shape of an NTC curve. At cool temperatures the resistance is high and changes rapidly with temperature, while at higher temperatures the resistance decreases more slowly.. The datasheet for the part provides the standard resistance versus temperature curve showing this characteristic exponential decay.

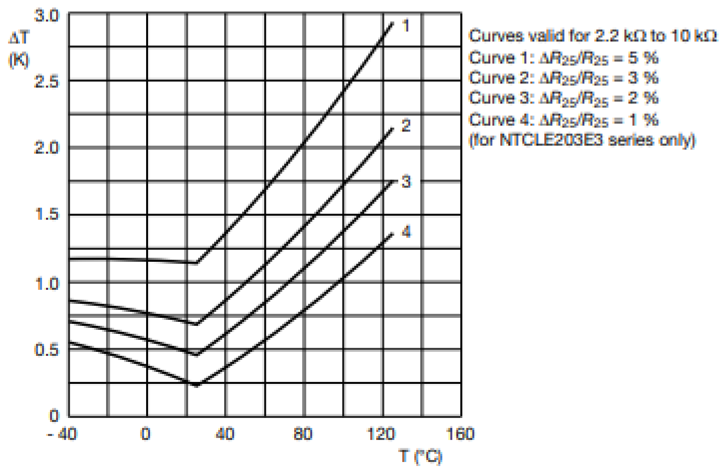


Figure: Temperature Deviation as a Function of the ambient temperature (from datasheet).

- Mathematic Model Chosen (pros/cons)
  - To convert resistance to temperature, the project uses the Beta model, a common single-parameter approximation:
  - $1/T = 1/T0 + (1/\beta) * ln(R/R0)$
  - This allows the thermistor resistance to be converted to temperature in Celsius using a simple algebraic expression. The model is computationally inexpensive and runs easily on the Pico.
  - What the model does not include:
    - Self heating effects in the thermistor
    - Nonidealities outside approximately 0 to 70 degrees Celsius
    - Aging effects and tolerance variations
    - Transient heating or cooling dynamics
  - Pros:
    - Simple algebraic form
    - Accurate enough for the temperature ranges relevant to this project
    - Only requires R- and Beta from the datasheet
  - Cons:
    - Less accurate at extreme temperatures
    - Does not capture all curvature in high precision applications
    - Assumes the thermistor follows the idealized Beta curve
  - Mode Range:
    - The Beta model is valid for the operating range of the reaction time station, which is indoor temperatures from roughly 18 to 30 degrees Celsius. For this project, the thermistor was only needed to detect overheating above approximately 95 degrees Fahrenheit, which is well within the stable region of the model

- Transducers
    - The thermistor itself forms one leg of a Wheatstone bridge. The bridge converts the resistance change into a small voltage change at the midpoint node. This midpoint voltage is then amplified by the INA121 instrumentation amplifier. The INA121 shifts and scales the differential signal so that the Pico ADC can read it as a clean single-ended voltage. This is the electronic transduction step: resistance becomes voltage through the bridge and amplifier.
- Signal Processing Options
    - 1. Direct voltage divider into ADC
        - Not acceptable, because the ADC resolution is not sufficient and the signal sits too close to the Pico noise floor.
    - 2. Wheatstone bridge without amplifier
        - Better linearity but still too small a voltage swing for reliable ADC readings
    - 3. Wheatstone bridge followed by an instrumentation amplifier
        - Provides strong common-mode rejection, precise differential gain, and the ability to bias the output around mid-rail for single-supply operation.
    - This third approach best meets the requirement that the signal processing be active and stable under environmental variations
- Power Options
    - Option 1: USB power only
        - Reliable but violates the project requirement of battery operation.
    - Option 2: Single 9V battery with regulators
        - Works for many sensors, but not enough current for the microcontroller plus WiFi.
    - Option 3: Dual battery source with 3.3V and 5V regulators
        - One battery for the thermistor conditioning and one for the Pico WiFi subsystem.
    - The final design uses the third option, with the instructor’s approval.
- Final Signal Processing Chosen
    - The chosen signal chain is:
        - 10k thermistor in Wheatstone bridge to
        - INA121 instrumentation amplifier to
        - RC ant-alias filter to
        - Pico ADC1 (GP27)
    - Key shaping performed by this circuit:
        - The bridge linearizes the thermistor response around the operating point
        - The INA121 applies a fixed gain of approximately 6
        - The RC filter reduces high frequency noise before ADC sampling
        - The ADC converts the amplified signal to counts for use in the Beta model

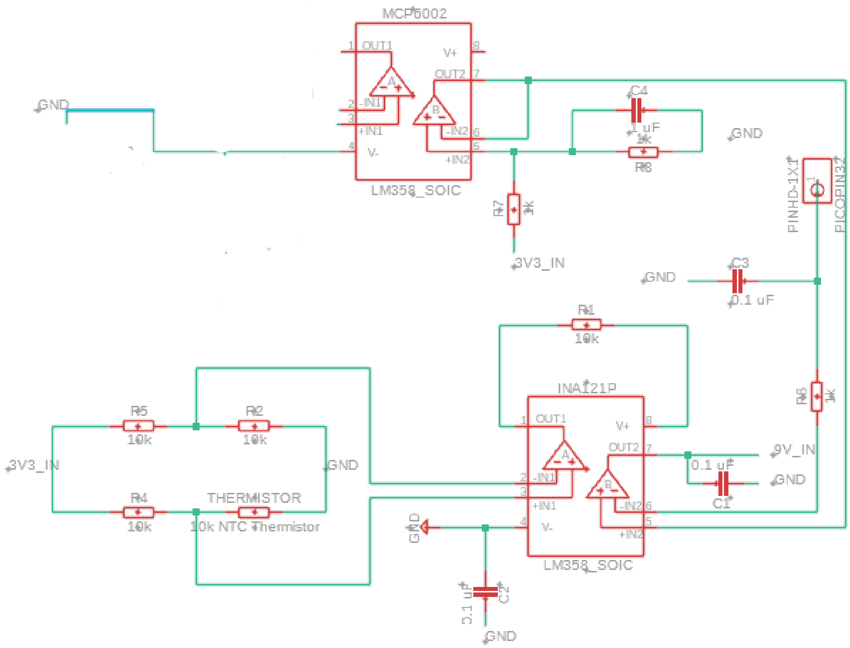


Figure: Circuit Diagram for Thermistor.

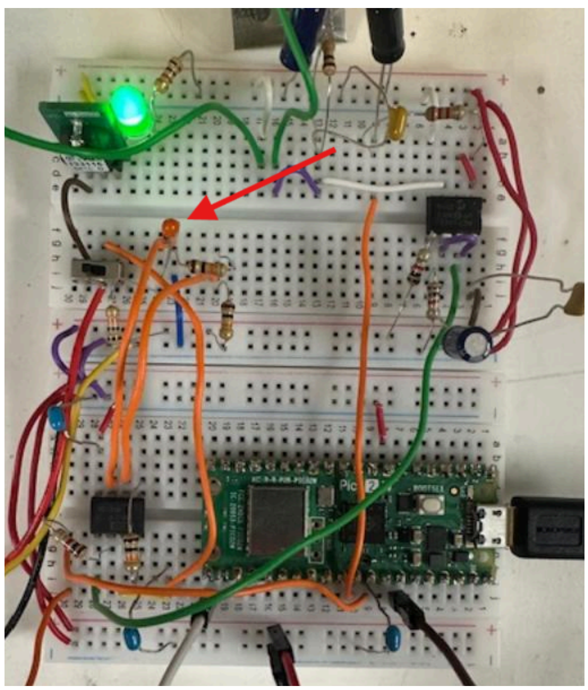


Figure: Physical Circuit for Thermistor.

- Filtering Chosen
    - The thermistor signal contains noise from:
        - ADC quantization

    - INA121 input noise
    - Digital switching noise from the Pico
    - WiFi current spikes coupling into the analog rails
  - An RC low-pass filter at the amplifier output helps remove high-frequency noise and stabilizes the ADC reading.
    - R = 1k ohms
    - C = 0.1 uF
  - The thermistor subfunction also uses a small digital moving average across 8 samples. This combination is enough to achieve a stable and repeatable temperature reading without adding latency that affects the protective over-temperature cutout.
- Microcontroller Code
  - The thermistor is encapsulated as a subfunction that performs the entire signal processing chain:
    - 1. Reads ADC counts
    - 2. Convert counts to INA output voltage
    - 3. Back-solve for the bridge midpoint voltage
    - 4. Compute thermistor resistance
    - 5. Compute temperature using the Beta model
    - 6. Apply optional calibration offset
    - 7. Return both Celsius and Fahrenheit
  - Calibration is done against R0 = 10k at 25 degrees Celsius, and a user adjustable offset can match the reading to a known thermometer. The 12-bit effective resolution of the Pico ADC gives more than enough accuracy given the narrow range needed for thermal safety monitoring.
  - The code for testing the thermistor in isolation can be found via the link below. *(Note: This was incorporated in the final system through its own function call)* https://drive.google.com/file/d/1gA6b8ZCaAPk7-HkLo7TRJ8N0Yqu3eaJ8/view?usp=sharing
- Kalman Filter Used
  - Unlike the FSR, the thermistor does not require high-speed filtering and does not experience rapid changes during the game. The thermistor’s moving average filter is sufficient. A Kalman filter was not used here because temperature changes are slow and the Beta model processing already smooths the signal. The raw data proves that noise after RC filtering and averaging is already within acceptable limits.
- Video Results
  - A quick demo showing the thermistor working in isolation can be viewed through the link below:

https://drive.google.com/file/d/1AuLfO9oEp3hH2J0kj5hZgSHdWyd3ycfx/view?usp=sharing

Other Sensors Used

Infrared Reflective Sensor (TCRT5000 Module)

- Raw Physics
  - The pre-packaged IR presence detector used in this project is based on the Vishay TCRT5000 reflective optical sensor, which integrates:
    - An infrared LED emitter
    - A phototransistor that receives reflected IR light from nearby surfaces
    - A plastic housing that filters visible light to reduce ambient interference
  - When an object is placed close to the sensor (typically within 0.2-15 mm), more IR light reflects back into the phototransistor, increasing its collector current. When the object is absent, reflection falls sharply.
  - Key Physics:
    - Signal depends on IR reflection, which depends on surface color, texture, distance, and angle.
    - Black and matte surfaces reflect poorly; white surfaces reflect strongly.
    - Maximum sensitivity occurs at ~2.5 mm from the sensor
  - The datasheet graph on page 4 shows the “Relative collector Current vs Distance” curve, confirming the typical peak at ~ 2.5 mm and monotonic decay beyond that distance

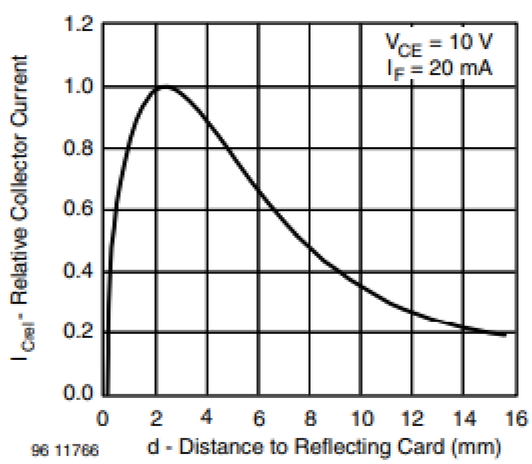


Figure: Relative collector Current vs Distance (from datasheet).

- Likely Internal Signal-Processing Path of the Module
  - The Raw TCRT5000 sensor provides only an LED + phototransistor pair. The purchased module version adds onboard circuitry:
    - 1. Current limiting resistor for the IR LED (fixed)

        - 2. Phototransistor load resistor converting light intensity into a measurable voltage
        - 3. LM393 comparator generating the digital DO signal
            - Threshold adjustable via on-board potentiometer
            - Comparator output is open-collector, pulled up through a resistor
        - 4. Optional AO analog output taken directly from the phototransistor divider
    - Thus the internal signal path is:
        - IR LED -> reflection -> phototransistor current -> voltage divider -> LM393 comparator -> digital output (DO)
- Output Characteristics
    - Digital Output (DO):
        - Active-LOW when an object is detected (phototransistor current exceeds threshold)
        - Range: essentially binary 0/1
        - Switching distance: adjustable via on-board potentiometer
        - Comparator hysteresis provides noise immunity, especially against ambient flicker
    - Analog Output (AO):
        - Continuous voltage from 0-3.3 V proportional to reflected IR intensity.
        - Useful for tuning threshold or analyzing reflectivity
        - Resolution depends on the Pico ADC
    - Datasheet characteristics (raw sensor):
        - Relative collector current > 20% from 0.2 mm to 15 mm
        - Peak detection distance ~2.5 mm
        - IR wavelength = 950 nm
        - Illumination is daylight filtered, reducing false triggering
    - The module's comparator introduces additional practical constraints, including threshold hysteresis and somewhat reduced usable range (~0-10 mm depending on the surface)
- Code (Library / Code used)
    - The project used MicroPython with no external libraries. The test script accessed:
        - Digital output via a GPIO input with internal pull-up
        - Optional analog output using the Pico ADC
    - Important functions:
        - read_detected(): reads digital DO line
        - read_ao_volts(): converts AO ADC reading to voltage
        - stable_detect(): 3-of-5 majority filter for noise robustness
    - This module required almost no calibration beyond setting the onboard potentiometer, making it ideal for the dual-sensor redundancy requirement.

- ○ The code used to test the IR in isolation can be seen via the link below.
  (Note: This was incorporated in the final system through its own function call)
  https://drive.google.com/file/d/1QfW7T_NXZyNEZS0xJ0H37lI8hAFW5_Pt/view?usp=sharing
- Demo Video
  - ○ A quick demo video showing the IR sensor working in isolation can be viewed via the link below:
  - ○ https://drive.google.com/file/d/1plkL4N3rdVboHBpME-F0ojqJ7oGO7-I-/view?usp=sharing

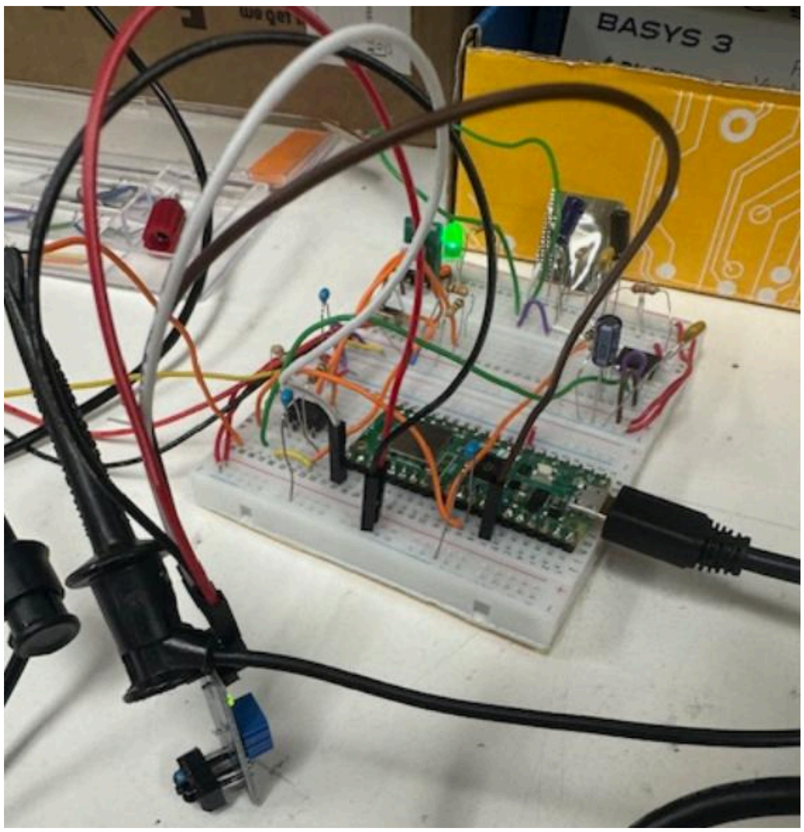


Figure: Physical Setup for the IR sensor.

Digital Logic Implementation

Minimum microcontroller requirements

To realize this reaction-time trainer, the microcontroller needed to satisfy a specific set of hardware and firmware requirements:

- Analog inputs
  - ○ At least two ADC channels with 12-bit resolution or betterL
    - ■ One channel for the FSR force sensor.

        - One channel for the thermistor / instrumentation-amplifier output.
- Digital GPIO
    - At least three digital inputs/outputs
        - One digital input for the TCRT5000 IR “presence” signal.
        - One digital output to drive the main game LED.
        - Additional GPIO reserved for debugging or future LED indicators
- Serial Buses
    - A hardware I2C controller to communicate with the SSD1306 OLED display at 400 kHz.
- Networking
    - On-board WiFi with TCP/IP support in MicroPython so that reaction-time scores can be posted to and retrieved from ThingSpeak.
- Compute and memory
    - Sufficient flash and RAM to run:
        - The raw-sensor math for the FSR and thermistor
        - A Kalman filter and moving average filter
        - A finite-state-machine (FSM) game loop.
        - Socket-based HTTP client code for ThingSpeak
- Power:
    - Ability to run directly from a regulated 5V rail (through VSYS) with battery power and to boot automatically into the main program.

Justification for using the Raspberry Pi Pico 2W

The Raspberry Pi Pico 2W satisfies all owthese requirements and is a good match for this project:

- It provides three 12-bit ADC channels on GPIO 26-28, which are used for the FSR and thermistor measurements.
- It has built-in 2.4 GHz WiFi, which removes the need for an external networking module and simplifies the IoT leaderboard.
- The chip exposes hardware I2C used to drive the OLED display at 400 kHz.
- The board offers plenty of GPIO pins for the LED, IR sensor, and future expansion.
- MicroPython support is mature, so the Kalman filter, Beta-model temperature math, and HTTP client can be coded and debugged quickly.
- The board can boot from battery-derived 5V on VSYS and automatically executes any file named *main.py.* For this project the final game logic is saved as *main.py* so that the reaction-time trainer runs as soon as the Pico is powered on, without a laptop.

Overall, the Pico 2W provided the right balance of analog capability, networking, cost, and ease of development for an interactive game that also includes an IoT leaderboard.

Overall software architecture

The final program is implemented in a single script, *main.py*, which imports a few helper modules:

- *ssd1306.py* for the OLED display driver
- *Wifi_connect.py* for a simple WiFi connection helper.
- *Thingspeak_post.py* for posting a reaction time to ThingSpeak.
- *secrets.py* for WiFi SSID/password and ThingSpeak channel keys

At a high level, the software pipeline is:

1. Raw sensing
    - FSR voltage is read via ADC0, smoothed by oversampling and a 1D Kalman filter, then converted to force using the piecewise-linear calibration derived earlier.
    - Thermistor voltage from the INA121 output is read via ADC1, converted back to bridge midpoint voltage, then to resistance, then to temperature using the Beta model.
    - The IR module's digital output is sampled with a small majority vote filter to reject chatter.
2. Safety and game logic
    - The current temperature is checked against a 95 degree fahrenheit limit on every loop. If exceeded, the state machine falls into a latched error state.
    - The FSR and IR signals feed a finite state machine that implements arming, random delay, anti-cheat, LED-on timing, and reaction-time measurement.
3. User interface
    - The OLED displays messages based on the current state (prompt to start, "Game Started, wait for LED," "LED ON," numeric results, and the Top-3 leaderboard).
    - The LED is controlled as a digital GPIO output and indicates when the user should react.
4. IoT logging and leaderboard
    - After each valid trial, the measured reaction time is posted to ThingSpeak.
    - After this post, the program waits for the ThingSpeak feed to refresh, then fetches the latest data and computes the current Top-3 times, which are displayed on the OLED.

Game state-machine algorithm

The core game behavior is encoded as a finite state machine with the following states:

1. WAIT_ARM
    - Default idle state.
    - OLED shows "Press button to start."
    - The code waits until both the FSR force exceeds the press threshold and the IR sensor sees an object. This ensures that the user has their hand in place.
2. WAIT_RELEASE

- OLED prompts “Release to begin.”
- The system waits until both the FSR and IR indicate that the button has been fully released. This avoids starting the timing while the user is already pressing.

3. PRIMED
   - The game has started but the LED is still off.
   - A random delay between 3 and 8 seconds is generated.
   - The OLED says “Game started, wait for LED, no touching.”
   - If the user touches the sensor early (FSR or IR active), the system transitions into the anti-cheat state.
4. ANTICHEAT
   - OLED displays “Don’t cheat, wait for LED.”
   - The system pauses briefly and then returns to the PRIMED state with a new random delay, forcing the user to restart the waiting phase.
5. LED_ON
   - When the random delay expires, the LED turns on and the OLED says “LED ON, press the button.”
   - The program records the first time the FSR trips and the first time the IR sensor sees the object again.
   - The reaction time is computed as the average of the FSR and IR delays from the LED-on moment, which helps cancel out timing jitter from either sensor.
6. SHOW_RESULT
   - After the dual-sensor event is detected, the code enters a short window to capture the peak FSR force.
   - The OLED shows the measured reaction time in milliseconds and the peak force in newtons.
   - In the background, the reaction time is posted to ThingSpeak once, and the time of the upload is recorded.
7. SHOW_TOP3
   - Once enough time has passed for ThingSpeak to refresh, the script downloads the latest feed, extracts field 1 values (reaction times), sorts them, and takes the three smallest.
   - Those Top-3 times are drawn on the OLED for a fixed duration, after which the system returns to WAIT_ARM
8. ERROR
   - If at any time the measured temperature exceeds the safety limit, the system enters a latched error state.
   - The OLED shows an over-temperature message and instructs the user to power down and let the system cool. The loop is effectively halted until the board is reset.

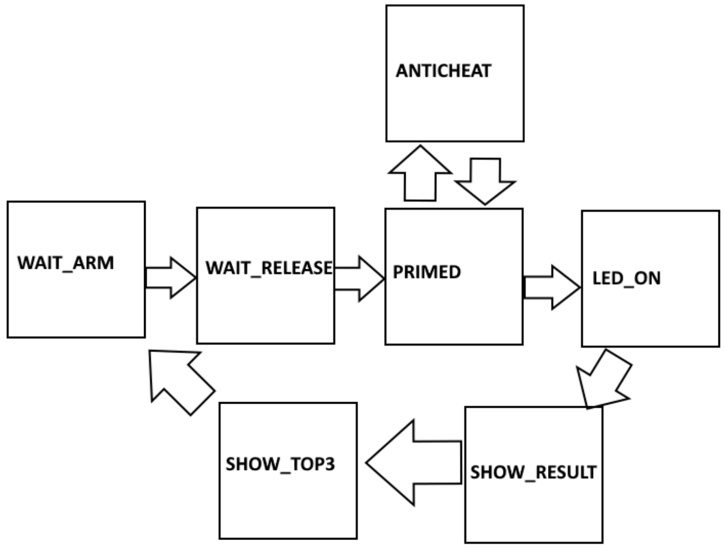


Figure: Quick Visual Representation of FSM.

IoT and leaderboard algorithms

The IoT functionality builds on the raw networking support of the Pico 2W:

- WiFi connection management
    - A helper function *ensure_wifi_connected()* checks if the WiFi interface is already associated.
    - If not, it displays "Connecting WiFi…" on the OLED, calls *wlan.connect()* with the SSID and password from *secrets.py*, and waits up to a fixed timeout.
    - This function is used both before posting and before fetching to avoid repeated connection logic in the main loop.
- Posting reaction times to ThingSpeak
    - *ts_post_reaction_ms()* is called once per trial from the SHOW_RESULT state.
    - It rounds the reaction time to an integer number of milliseconds and passes it to *thingspeak_post.update()*, which constructs an HTTP POST with the write key and puts the value into field 1.
    - The function returns a success flag so that the main state machine only proceeds to the leaderboard fetch once the upload succeeds.
- Fetching and displaying the leaderboard
    - After a successful upload and a wait of roughly 20 seconds (to respect ThingSpeak rate limits), the code calls *ts_fetch_top3_ms()*.

- This function opens a TCP socket, sends an HTTP GET request to the ThingSpeak API for the configured channel and field, and receives the JSON response.
- The JSON is parsed to extract all numeric values of field1. These are converted to floats, sorted, and the three fastest times are kept in a cached list.
- The OLED renderer *oled_top3_ms()* displays these three times with ranks 1-3, or “No data yet” if the feed is empty.

Together, these algorithms turn each play into both a local result and a contribution to a shared, persistent leaderboard.

The final *main.py* script used for the full pipeline can be viewed via the link below:

https://drive.google.com/file/d/1kCJafePH6QgQRmSlC0Lmg7NtEmIG3E_j/view?usp=sharing

Final Design and Experimental Results

Final Hardware Overview

The final reaction-time trainer integrates all raw sensor conditioning, digital logic, power management, and user interface components into a compact, enclosed system. The system includes:

- One force sensor (FSR) used as a physical “button”
- A TCRT5000 IR presence detector providing redundant detection
- A Wheatstone-bridge thermistor circuit amplified through an INA121 in-amp
- A 128x64 OLED for UI output
- A Raspberry Pi Pico 2W running the entire state machine and Wi-Fi leaderboards
- A custom dual-battery power system with PMOS switching
- A designed PCB holding all raw-signal analog stages (not manufactured)

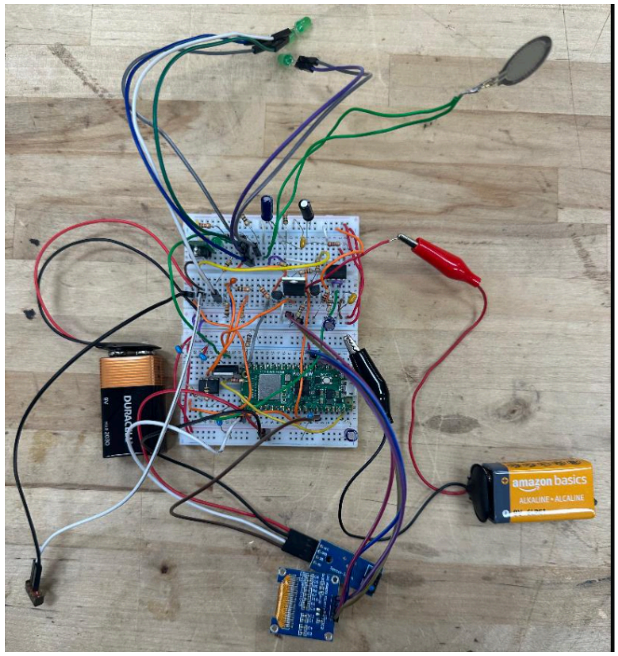
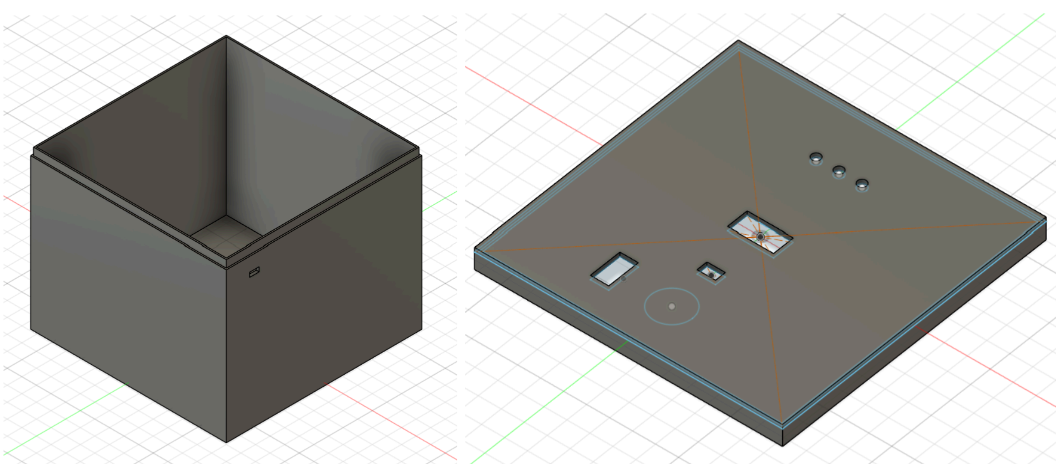


Figure: Finalized breadboard circuitry.

3D Printed Enclosure

The enclosure was designed in Fusion 360 and printed in PLA. The design focused on mechanical stability, ease of component placement, and clean user interaction. The lid includes all openings sized for the sensors and display:

- Three LED holes
- Rectangular cutout for the OLED display
- Square cutout for the FSR and IR sensors
- A small rear port allowing USB access to the Pico during development
- Mechanical lip ensuring snug, repeatable lid alignment



Figures: CAD Models for 3D printed Enclosure.

Figure: Printed Box with Components Placed.

Power System and Battery Setup

The final design uses a dual-battery arrangement to reliably power both analog and digital subsystems. A single SPST switch controls the entire system by simultaneously:

1. Enabling the 3,3V linear regulator supplying the analog front end
2. Driving the gate of a PMOS high-side switch that enables the second 9V battery feeding an LM317-based 5V regulator for the digital logic and Wi-Fi subsystem

A small NPN stage isolates the gate drive so both power rails activate cleanly from the single toggle. Bulk capacitance was added to prevent Wi-Fi surge dips from collapsing VSYS.

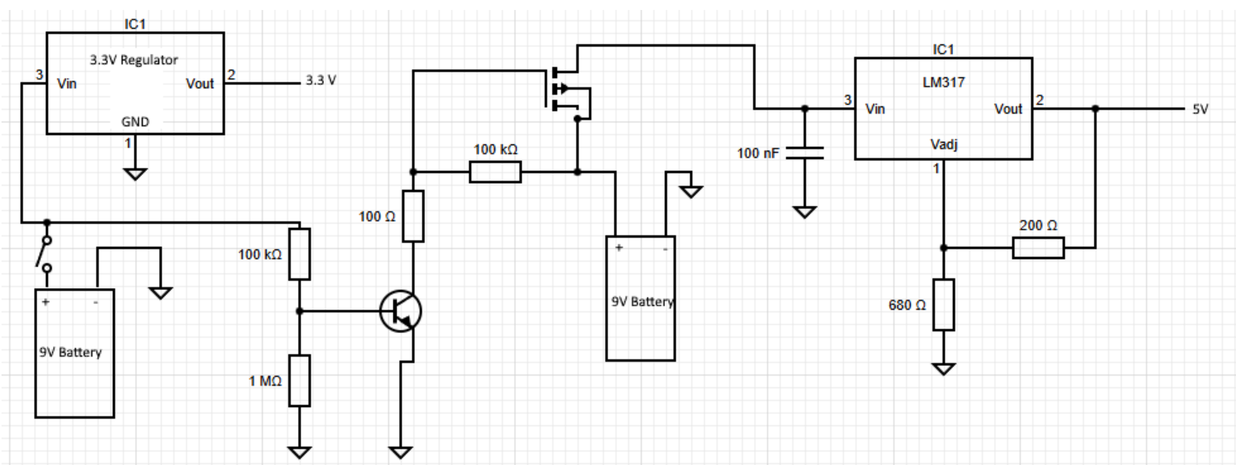


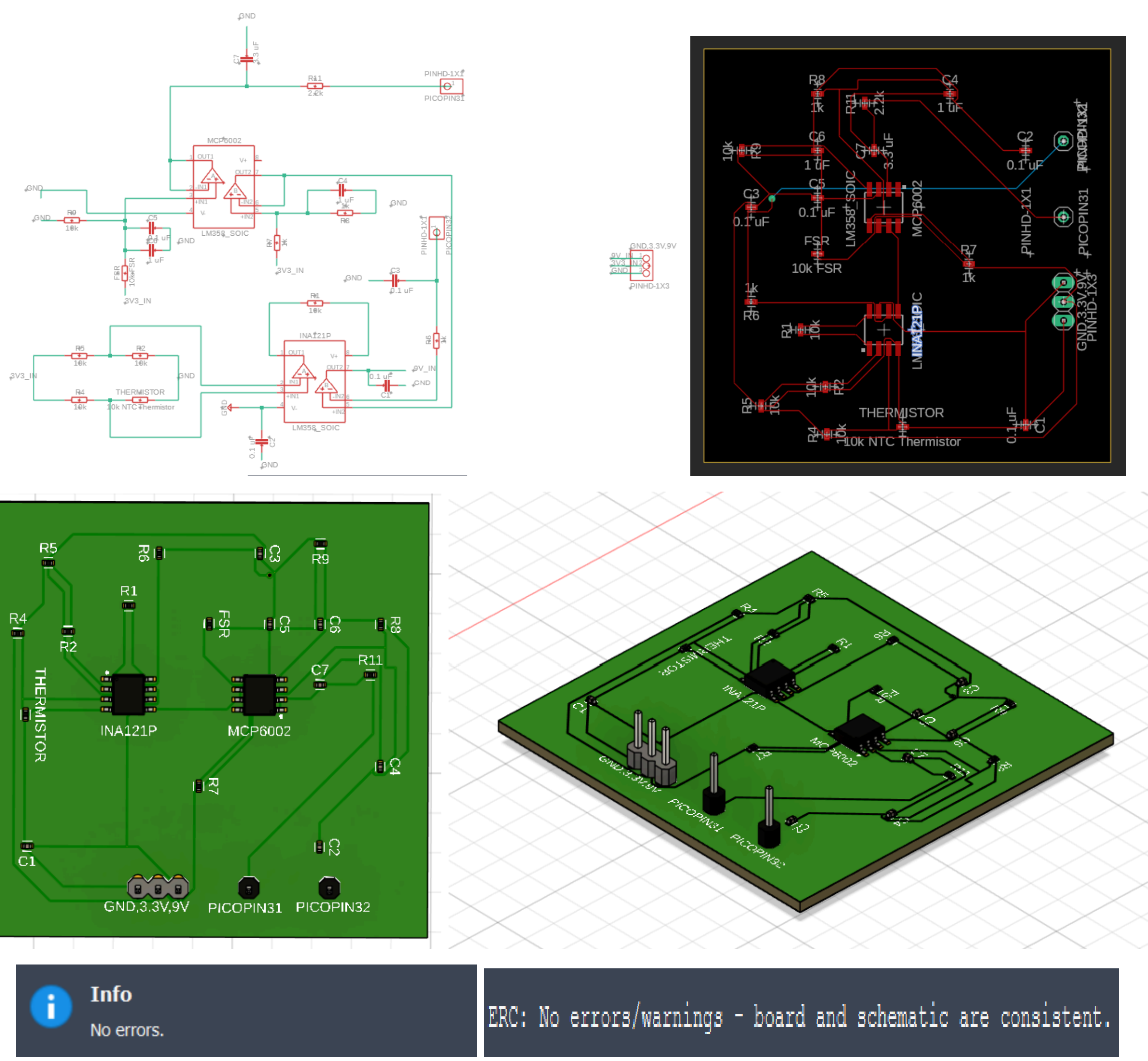
Figure: Circuit Schematic of For the Power System.

PCB Design

Since raw signal processing occupied the most breadboard space, the PCB was designed to move all analog filtering, amplification, and bridge circuitry off the breadboard. This includes:

- INA121 Wheatstone-bridge instrumentation amplifier for the thermistor
- MCP6002 op-amp buffering and filtering for the FSR
- MCP6002 op-amp for smoothing and conditioning
- Input/output headers for Pico pins, 3.3V, and ground
- All recommended bypassing and anti-alias filtering components
- Clean ground routing with separation between analog and digital returns

ERC and DRC checks were completed and passed.



Figures: Images showing the Process of the PCB design. From Schematic to 3D rendering. Also the output from running the DRC and ERC.

Although the PCB was fully designed with manufacturable routing, it was not fabricated due to time constraints. However, all ERC/DRC checks passed, confirming that the design is electrically valid.

Intermediate Signal-Processing Results

Because this project includes two raw sensors, intermediate voltage levels were measured and verified at the following locations:

- FSR voltage divider output
- FSR low-pass buffer output
- Thermistor bridge midpoint V1
- INA121 amplifier output
- Filtered signals going into ADC0 and ADC1 on the Pico

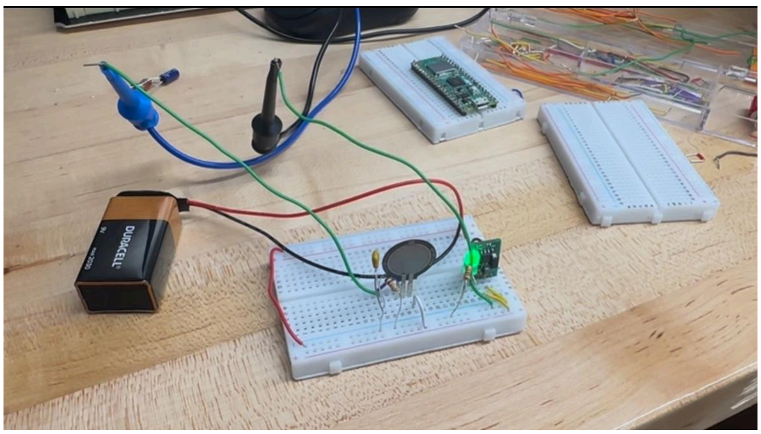


Figure: Example showing early debugging with FSR.

Experimental Results

The final system was assembled, enclosed, and tested under normal operating conditions. The final demonstration successfully executed all required functions:

- LED-triggered reaction timing
- Dual-sensor press detection (FSR + IR)
- Thermistor-based thermal shutdown
- OLED user interface
- Wi-Fi upload to ThingSpeak with automatic leaderboard display
- Battery-powered standalone operation

A consistent sequence of measurements was observed across repeated trials, and the system correctly detected press events, computed reaction time, measured peak force, and displayed results. ThingSpeak logs confirmed that IoT transmission worked as intended, and the professor-observed in-class demo validated the system’s real-world behavior.

During extended testing, two practical issues were identified:

1. Uneven LED brightness

  - Although all LEDs were functional, their brightness levels did not match. One LED consistently appeared dimmer than the others, and overall brightness was lower than desired for high-visibility cues. This mismatch likely resulted from resistor tolerances and the fact that all LEDs were sourced from differential batches. This did not affect functionality but did affect user perception.
2. Mechanical instability of the power switch wiring
  - The physical legs on the toggle switch were shorter than expected. Because of this, male-to-femail jumper wires sometimes made inconsistent contact. Under decent movement, the switch could intermittently disconnect, requiring the wires to be pushed further onto the legs. The problem was mechanical rather than electrical and did not occur once a stable connection was achieved, but it highlighted a manufacturability weakness in the prototype assembly.

Despite these minor issues, the system performed reliably during the formal demonstration, and the overall functionality matched the design intent.
The **final demo video** showing the full pipeline can be viewed via the link below:
https://drive.google.com/file/d/13KiLQajpabt88BUegUfWbOQVhlgvolvL/view?usp=sharing

Analysis of the Results

From an engineering standpoint, the experimental results show that the sensing chain, digital logic, and IoT integration worked as designed. The system produced stable and repeatable measurements, the Kalman filtering improved raw signal quality, and the state machine responded predictably across all edge cases. While the PCB was not physically produced, its layout was intentionally designed with short traces, clean grounding, and proper routing that would significantly improve noise performance over the breadboard prototype.

The two observed shortcomings point to clear next-iteration improvements:

- LED brightness inconsistency can be fully corrected by selecting LEDs from the same batch, matching forward voltages, and using equal-value resistors or a constant-current LED driver. Increasing total current drive (while staying within GPIO limits) would improve overall visibility.
- Switch connection instability can be solved by either using a switch with longer solder-friendly terminals or by integrating the switch directly onto the PCB in future revisions. A locking JST connector or solder-lug terminals would eliminate intermittent contact entirely and improve robustness.

These issues are normal for a first hardware iteration and do not affect the core performance or validity of the measurement system. Instead, they simply highlight mechanical and ergonomic improvements that would be incorporated into a production-quality revision. Overall, the system satisfies the project requirements and demonstrates a complete, functional embodiment of sensor physics, analog signal conditioning, digital logic, and embedded IoT communication.

Conclusions

Improvements for a Revision 2

If I were to develop a second revision of this project, several improvements immediately come to mind, both in the hardware and in the overall system design. The most important change would be moving toward a fully soldered wiring solution. During testing, some jumper-wire connections, especially those attached to the small switch legs, became loose or required re-seating to function reliably. Hard-wiring the sensors or using proper crimped connectors would dramatically improve long-term robustness.

A second major improvement would be fabricating the PCB that I designed. The breadboard version worked, but it occupied a significant amount of space and introduced noise and mechanical fragility that a PCB would eliminate. In a Revision 2, I would either print the existing PCB or expand it into a multi-functional board that includes additional components such as the LED driver, power-conditioning circuitry, and connectors for the OLED and IR sensor. With more components integrated directly onto the board, the enclosure could be made smaller, cleaner, and more reliable.

The enclosure itself also has room for refinement. The 3D-printed box served its purpose, but a more compact and ergonomic design would enhance the overall experience. Integration of the printed PCB into the enclosure would allow for a sleeker footprint and better internal cable management. In Revision 2, I would also enlarge the OLED cutout and use a larger display. The current screen works, but a bigger OLED would make the feedback easier to read and would make the game experience more polished.

The LEDs used in the project were bright but not uniform. Two were noticeably dimmer than the third, which distracted from the user experience. A proper LED driver or matched-resistor network would solve this. I would also consider using a single high-intensity RGB LED.

On the software side, the IoT upload speed was restricted by ThingSpeak’s 15-second minimum update interval. If I were to revise this project, I would explore alternative cloud platforms or build a lightweight backend hosted on a personal server, allowing faster leaderboard updates. This would make the interactive portion of the system feel more immediate.

Finally, I originally intended to incorporate a physical arcade-style button. Mechanical mounting proved time-consuming, and with time constraints from other classes, I chose not to force it into the design. In a future revision, I would prioritize a proper arcade button, since it would dramatically improve the tactile feel of the project.

If I was Assigned This Again

If this assignment were repeated, there are several changes I would hope for in the structure and timeline. First, I would appreciate earlier confirmation on power-supply exceptions. My design required dual battery sources due to Wi-Fi load on the Pico 2W, and receiving that approval earlier would have simplified planning. More guided recommendations for power-switching hardware (such as MOSFET selection) could reduce the trial-and-error needed to design a safe and reliable power system.

I would also suggest that future iterations of the assignment encourage students to fabricate their PCBs earlier in the semester, or provide optional PCB workshops so students can transition from breadboards to manufactured boards more effectively. This would align well with the course's emphasis on robust sensor design and would allow more time to debug the PCB in the final build.

Finally, structured intermediate deadlines for enclosure design, PCB routing, and code integration would help distribute workload more evenly across the semester. This project required many interconnected pieces, and breaking those into formal checkpoints would benefit future students.

Lessons Learned

This project pushed me deeper into practical sensor integration than any previous coursework. I learned how important it is to balance theory with physical constraints like noise, wiring reliability, mechanical tolerances, and power stability, which all played larger roles than I initially expected. I also learned that even simple mistakes, like loose switch wiring, can strongly affect usability, and that careful attention to these details is essential.

I gained experience working with multiple sensors simultaneously, each using different physics and different signal-processing pipelines. Writing clean subfunctions for each sensor forced me to think modularly, and implementing the Kalman filters showed me firsthand how much software can improve raw analog measurements.

The project also reinforced the value of planning for mechanical design early. My enclosure worked, but I underestimated how much time mechanical fitment, hole sizing, and wiring clearances would require. Next time, I would prototype the enclosure sooner and build the electronics around the mechanical design, not the other way around.

Overall, this project taught me how to think like a systems engineer rather than just a programmer or circuit designer. Bringing together analog signal conditioning, digital logic, IoT communication, mechanical CAD, PCB layout, and user experience into one cohesive device was challenging, but it was also the most rewarding part of the semester.

Human References

Datasheets

All of the datasheets can be found through the link below:
https://drive.google.com/drive/folders/1ZexaMLxrxzexjHxvciaTV4B7kMsLvEbV?usp=sharing

Resources Used

- For help creating a PCB: https://www.youtube.com/watch?v=eEdnImVezi8
- Fusion: https://www.autodesk.com/products/fusion-360/overview
- Schematic Builder: https://www.circuit-diagram.org/

AI References

This project used AI tools to support documentation, circuit verification, code debugging, and general technical explanations. AI assistance was used only for clarification, error-checking, and writing support. While most circuit design, coding, testing, and physical construction were completed by me.

Below is a list of example prompts that reflect the type of AI assistance used during development.

Example Prompt 1 - Report Structure and Outline

"Can you create an outline for the sections of this sensors project report so I can understand how to structure the document?"

Example Prompt 2 - Component Selection and Circuit Values

"Given this operational amplifier circuit with an FSR and a 10k resistor, what resistor and capacitor values should I choose to create a stable gain and filtering stage?"

Example Prompt 3 - Code Debugging

"My MicroPython script crashes when reading ADC values and converting to voltage. Can you help identify what line is causing the error and how to fix it?"

Example Prompt 4 - Explanation of Sensor Physics

"Can you explain the raw physics behind a 10k NTC thermistor and how the Beta model converts resistance to temperature?"

Example Prompt 5 - Documentation Support

"I need to write the Mathematical Model section for my FSR. Can you summarize how the voltage divider equation converts applied force into measurable voltage?"

Example Prompt 6 - Troubleshooting Hardware Behavior

"My IR sensor module sometimes triggers randomly. What wiring or environmental factors could cause false positives?"

Appendix

Appendix A - Complete Bill of Materials (BOM)

Component	Part Number	Vendor	Cost (USD)	Quantity	Link
Raspberry Pi Pico 2W (RP2350 + WiFi)	SC0915 (Pico 2 W)	Amazon	12.99	1	https://a.co/d/ao0QkTz
3.3V Step-down Regulator	OKI-78SR-3.3/1.5-W36-C	Digi-Key	4.97	1	https://www.digikey.com/en/products/detail/murata-power-solutions-inc/OKI-78SR-3-3-1-5-W36-C/2259780
9V Alkaline Battery	MN1604	Amazon	8.97	1	https://a.co/d/2SYFJ7D
9V Battery Snap Connector	36-233-ND	Digi-Key	0.54	1	https://www.digikey.com/en/products/detail/keystone-electronics/233/68726
Slide Switch	CKN12284-ND	Digi-Key	0.76	1	https://www.digikey.com/en/products/detail/c-k/OS102011MS2QN1C/10063871?s=N4IgTCBcDaIMIDIDSACA8gZQIwAYw6ywFkMwBFAOSzhAF0BfIA
Force Sensing Resistor (FSR)	1027-1018-ND	Digi-Key	8.9	2	https://www.digikey.com/en/products/detail/interlink-electronics/34-00015/5416350
5V Regulator	LM317	Digi-Key	0.77	1	https://www.digikey.com/en/products/detail/onsemi/LM317TG/918508
10k NTC Thermistor	BC2395TR-ND	Digi-Key	1.5	2	https://www.digikey.com/en/products/detail/vishay-beyschlag-draloric-bc-components/NTCLE100E3103HT1/2230723
IR Sensor	TCRT5000	Amazon	5.99	1	https://a.co/d/hgrk7V5
Dual RRIO Op-Amp (DIP-8)	MCP6002-I/P-ND	Digi-Key	0.44	2	https://www.digikey.com/en/products/detail/microchip-technology/MCP6002-I-P/500875?s=N4IgTCBcDaILIGEAKA2ADGshEAgJIHokQBdAXyA
Perfboard / Perma-Proto Half-size	Half-size perfboard	Amazon	6.99	1	https://a.co/d/c5IDrlJ
Resistor Assortment 1/4W 1% (1Ω–1MΩ)	Assortment Kit	Amazon	9.99	1	https://a.co/d/12S78or
Ceramic Capacitor Assortment (50V, 10pF–100nF)	Assortment Kit	Amazon	12.99	1	https://a.co/d/dVBR6DY
Electrolytic Capacitor Assortment (10–470µF)	Assortment Kit	Amazon	16.99	1	https://a.co/d/cDJyDn0
Jumper Wire Kit	ELEGOO M/M M/F F/F	Amazon	6.98	1	https://a.co/d/5VFRsiW
LED Assortment	5 mm diffused (multi-color)	Amazon	6.99	1	https://a.co/d/avANbkp
0.96" I2C OLED Display (SSD1306)	SSD1306 0.96" I2C	Amazon	9.99	1	https://a.co/d/40vCJPg
Solderless Breadboard	ELEGOO 830 Point Solderless Prototype PCB Board	Amazon	8.99	1	https://a.co/d/01NQj1R
Male & Female Header Strip 2.54mm (40-pin)	OCR 40PCS 2.54mm Male and Female header Connector	Amazon	10.99	2	https://a.co/d/8hXYqX0

Figure: Complete BOM for this Project.

This BOM is downloadable via the link below:
https://docs.google.com/spreadsheets/d/1Qei9-RZAJLz0pikmknKVU8e6-Psdinkl/edit?usp=sharing&ouid=118322779747190666804&rtpof=true&sd=true

Appendix B - General Safety Checklist

Sensor (and the environment)

Area	Check	Pass/Fail	Environment Range	Sensor Range	Comments
Environment*	Temperature Range	Pass	20-25 degrees celcius	FSR: 0-70 C Thermistor: -40 to 125 C IR: 0-50 C	Indoor temperatures are well within all sensor operating limits
	Max shock and vibration	Pass	Very low; device is stationary	All sensors tolerant to normal handling	No significant vibration sources presents
	Humidity	Pass	30-50 percent RH	All sensors non-sealed but fine for normal indoor humidity	No exposure to condensation
	Pressure	Pass	Standard atmospheric pressure	Not pressure sensitive	No pressure variation expected
	Acoustic Level	Pass	Normal conversation levels (~60 dB)	not acoustically sensitive	Not affected by sound
	Corrosive Gases	N/A	N/A	N/A	N/A
	Magnetic and RF Fields	Pass	Typical indoor EM environment	No magnetic sensitivity and only minor RF susceptibility	No nearby strong RF emitters; WiFi is exptected and harmless
	Nuclear Radiation	N/A	N/A	N/A	N/A
	Salt Spray	N/A	N/A	N/A	N/A
	Transient Temperatures	Pass	Very small, +/- 2 C variations	All sensors support wide temperature span	No rapid heating or cooling expected
	Strain in the Mounting Surface	Pass	Stable desktop, no flexing	FSR requires flat mounting Thermistor fixed in breadboard IR mounted to enclosure	Enclosure provides stable non-flexing mounting.

*Overall accuracy is MOST affected by sensors characteristics such as environmental effects and dynamic characteristics.

Area	Check	Pass/Fail	Environment Range	Cable Range	Comments
Sensor Cable*	Temperature Range	pass	20-25 deg celcius	(-20) - 80 deg celcius	Indoor temperatures do not stress cable insulation
	Humidity Conditions	Pass	30-50 percent RH	~90 percent RH	No risk of condensation; enclosure keeps cables protected
	Noise levels	pass	No significant electrical noise sources nearby	Short wire lengths minimize EMI pickup	ADC lines are short and routed inside enclosure, reducing susceptibility
	Size and weight	Pass	lightweight benchtop system	Jumper wires are low mass and easy to route	No strain on connectors; enclosure supports all cables
	Flexibility	Pass	System is stationary during use	Flexible silicone/PVC insulation	Only light movement during assembly; no repetitive bending.
	Is a sealed connection req?	No			

Sensor (FOR FSR)(is this the correct sensor?)

Area	Characteristics	Datasheet Value	At what temp/condition?	Comments
Sensor*	Sensitivity	High sensitivity to normal force; resistance changes by roughly an order of magnitude over 0.2–10 N	25 C static finger press on pad	Sensitive enough to clearly distinguish no touch, light touch, and firm press needed for the game button.
	Frequency Response	Specified for low-frequency to several kHz (quasi-static loads)	Room temperature, dynamic presses	More than fast enough for human reaction time events in the tens of milliseconds.
	Resonance Frequency	Not explicitly specified	N/A for this application	Device is a thin polymer film; no problematic resonance in the low-frequency range of human pressing.
	Minor Resonances	Not specified	N/A	Any minor mechanical ringing is negligible compared to digital sampling and Kalman filtering.
	Internal Capacitance	Not specified (small)	N/A	High input impedance of the op-amp and ADC makes internal capacitance insignificant.
	Transverse Sensitivity	Responds mainly to normal force; limited sensitivity to shear	Room temperature, finger press	Some off-axis loading possible, but sensor is mounted flat so transverse effects are acceptable.
	Amplitude and Linearity	Strongly non-linear, approximately log(force) vs log(resistance)	25 C	Nonlinearity handled with piecewise linear calibration in software; accuracy is adequate for relative force display.
	Hysteresis	Moderate; resistance on unloading differs from loading	Repeated presses at room temperature	Acceptable because game only needs approximate peak force, not precision metrology.
	Temperature Deviations	Resistance varies with temperature (not fully specified)	Room temperature indoor use	System is not intended for wide temperature swings; thermistor monitors enclosure temperature separately.
	Weight	Very low (flexible film, < 1 g)	N/A	Adds negligible mass to the enclosure; easy to mount under top plate.
	Size	Approx. coin-sized active area (small round FSR)	N/A	Fits comfortably on the lid with space for OLED and LEDs.
	Internal Ω at max Temp	Resistance decreases with force; nominal a few kΩ at high load	25 °C, near maximum usable force	Divider and op-amp circuit sized so ADC never saturates over the expected resistance range.
	Calibration Accuracy	Absolute accuracy is limited; device is intended for relative force measurement	Bench calibration using known masses at room temperature	Piecewise linear calibration gives reasonable repeatability for the game; not a precision load cell.
	Strain Sensitivity	Flexible polymer; can be damaged by sharp bending	Mechanical mounting on flat lid	Mounted on a flat, rigid surface to avoid extra strain unrelated to button pressing.

	Damping at Temp extremes	Not specified	N/A for indoor environment	Device is only used near room temperature, so temperature-dependent damping is not a concern.
	Zero Measurand Output	Finite resistance even with no load (tens to hundreds of kΩ)	No finger contact at room temperature	Circuit and software treat this baseline as "no press" and use thresholds above 1 N.
	Thermal Transient Response	Not specified	N/A	Thermal dynamics are slow compared to electrical sampling; effect is negligible for quick button presses.

*The most important element in a measurement system is the sensor. If the data is distorted or corrupted by the sensor, there is often little that

Figures: Completed First 4 Pages of General Safety Checklist.

Appendix C - FSR Characterization

Calibration and Characterization Procedure

To characterize the sensor, known masses were stacked on a 3D-printed "table" that sits atop the FSR, and the resulting output voltages were measured using the analog circuit described above. Each mass was converted to force using F = m * g (with g = 9.80665 m/s^2 and m in kg). The data collected are shown below:

Mass (g)	Vout (V)
0	0
107	1.25
207	1.9
307	2.4
407	2.5
507	2.75
607	2.85
707	2.9
807	2.93
907	2.95
1007	2.97

Figure: Data collected during characterization.

These data were used to derive a transfer function mapping measured voltage to applied force.

Transfer Function Derivation (Linearization)

Because FSRs exhibit a nonlinear resistance-force relationship, the raw Vout values were converted into a linearized force scale using a piecewise-linear interpolation method. Each consecutive pair of measured data points (Vi, Fi) and (Vi+1, Fi+1) defines a linear segment:

$$F(V) = F_i + \frac{(V - V_i)}{(V_{i+1} - V_i)} (F_{i+1} - F_i)$$

This approach divides the nonlinear curve into small linear regions, ensuring accurate mapping within the tested range without resorting to complex polynomial fits. The valid operating region was defined up to 2.9 V, beyond which the sensor begins to saturate.

All of this is talked about in more detail in the Checkpoint 3 Report:
https://docs.google.com/document/d/1u1aaE8vvv8TbaEjbHUW4ZGgO4YzIY1qfSoMuOg0u8tg/edit?usp=sharing